Scenario 1: Satellite Signal Eavesdropping

👋 **Hello, Hacker!** Today's mission: quietly grab the secret signal a satellite is beaming down. Follow the stages below in order and anyone can snoop on a satellite. Gear up and let's go.

Mission map (the flow)

> **INTEL → 1 TRACK → 2 ANALYZE → 3 PUZZLE → 4 EXECUTE → RESULT**

The step indicator at the top shows which stage you are on. Clear each stage's "To do" top to bottom, then move on.

Your mission

Become the attacker: eavesdrop on the downlink signal the satellite sends down and recover the image it carries.

Mission briefing (INTEL)

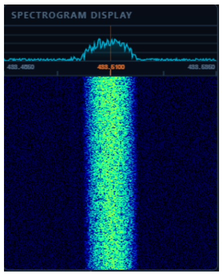
A satellite talks to its ground station over radio (RF), so match your receiver to the RF parameters it uses (frequency, modulation, and so on) and you receive the same signal. The values you need are in **SAT INFO** at the top.

STAGE 1: Catch the signal (TRACK)

On screen: Left is the orbit tracking, right is the 3D antenna view with the received-signal waterfall, and the control panel is at the bottom.

To do (in order):

1. **Reset / Time remaining:** If the satellite is too far or below the horizon, use Reset to jump to just before the next pass.
2. **Antenna:** Pick the antenna matching the satellite's polarization. Correct is green, wrong is red.
3. **Track satellite:** Turn on tracking so the antenna follows the satellite automatically.
4. **Center frequency:** Enter an appropriate center frequency and press Apply.
5. **Doppler correction:** Correct the frequency shift from the fast moving satellite.
6. **Bandwidth:** Widen the bandwidth so the signal is not clipped in the waterfall.
7. **Record:** Once the signal comes in well, record for a few seconds. This file feeds the next stage.



STAGE 2: Unmask the signal (ANALYZE)

To do: Upload the file from STAGE 1, then read the center frequency (peak in the spectrum), bandwidth (how wide the signal is), and modulation (two separate tones means FSK).

STAGE 3: Build the demod flowgraph (PUZZLE)

To do: Read each block's category (SOURCE → FILTER → DEMOD → DEFRAME → SINK) and drop it into the right slot. Correct turns green and the wiring connects, wrong turns red and stays disconnected.

STAGE 4: Run the demodulation (EXECUTE)

Check: Press Live decode and the image is rebuilt row by row. If the recorded signal's center frequency is off, the image comes out tilted.

MISSION CLEAR: The Recovered Image

Mission clear. With passive reception alone and no special intrusion, you recovered the very image the satellite sent down. Even ordinary, legitimate communications can be exposed to eavesdropping.